FR1 FR2 TEST TOOL USER GUIDE V4.9

System Engineering

DEC 2024

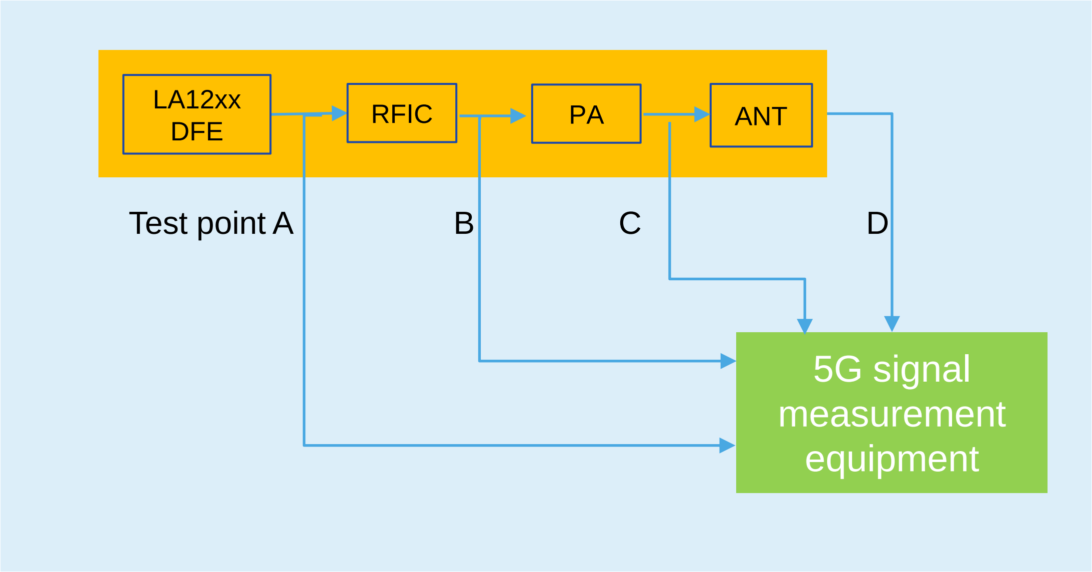
SECURE CONNECTIONS
FOR A SMARTER WORLD

CONTENT

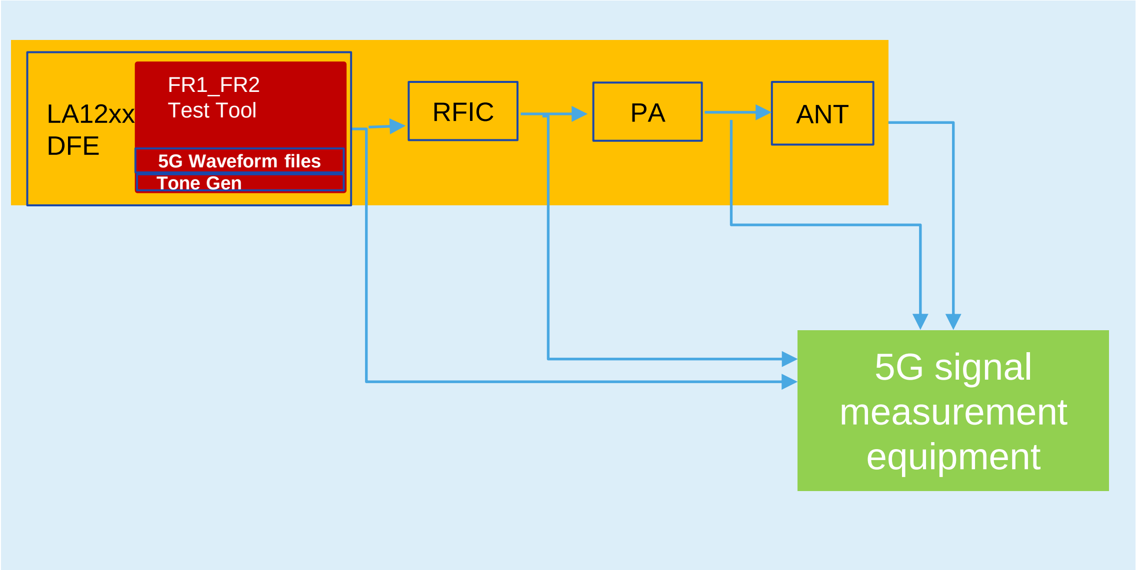
- What is FR1 FR2 Test Tool
- Test cases
- Test Tool functionalities
- Test Tool block diagram
- Test Tool package
- Test Tool commands
- Guidance for generating test vectors using Matlab

OVERVIEW: WHAT IS FR1 FR2 TEST TOOL

5G RF Product diagram



Test RF card with the Test Tool



All customers need a signal sent from LA12xx/LA93xx device to RF to test RFIC/PA/ANT:

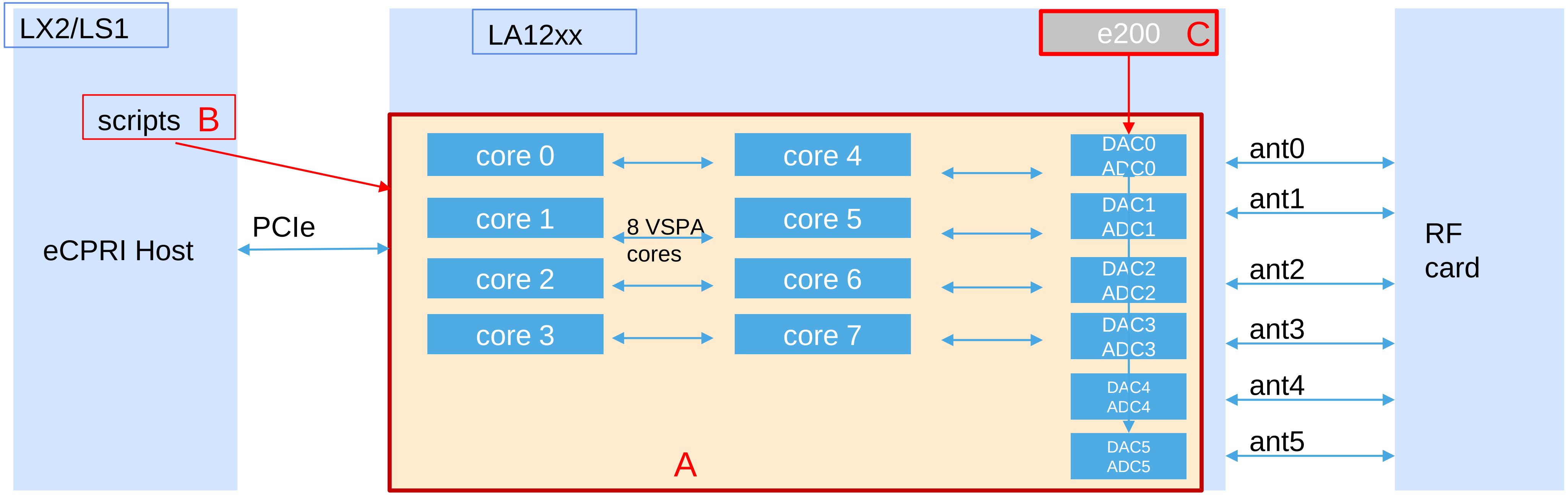
- RF board bring-up
- RF component calibration
- RF performance test

FR1 FR2 Test Tool is a software tool located in LA12xx/LA93xx device, which can send 5G waveforms to RF

OVERVIEW: WHAT IS FR1 FR2 TEST TOOL

- FR1_FR2 Test Tool is a software tool running on LA12xx device. Which can:
  - Generate 5G signals or single tone on LSDCS for sub-6G FR1 to test RF card performance
  - Generate 5G signals or single tone on HSDCS for mmwave FR2 to test RF card performance
  - Generate eCPRI symbol data sent over PCIe to eCPRI host for timing/performance test
    - Bandwidth supported currently:
      - 50Mhz 15Khz SCS
      - 100Mhz 30KHzSCS
      - 400Mhz 120KHzSCS

FR1_FR2 test tool = A (vspa images to generate 5G signal) + B (host scripts to control vspa) + C (e200 software to control DCS)



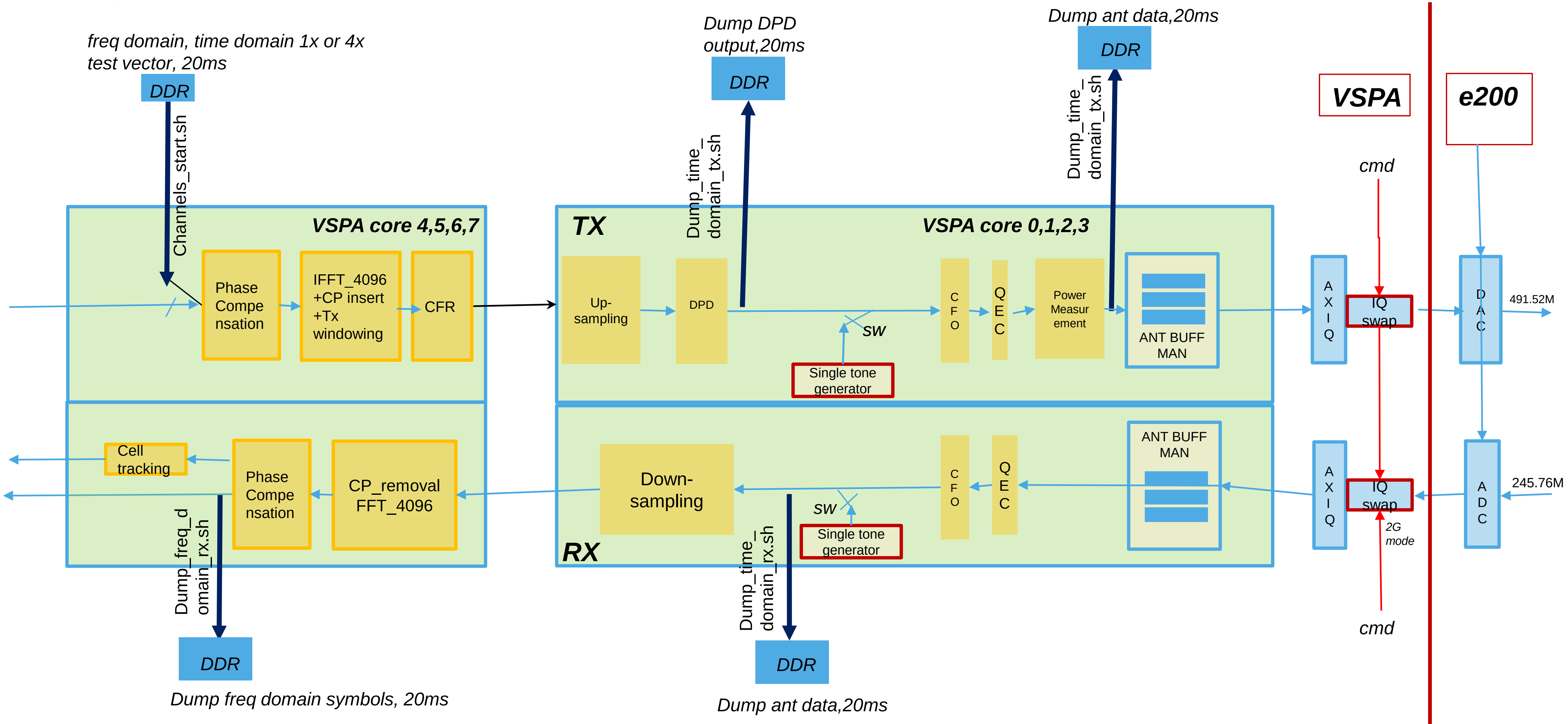
TEST CASES

- TX basic functional test
  - Send single tone or 5G TM signal, test each part of the RF transmit path.
- TX QEC test
  - Based on basic functional test result, do QEC calibration to get IQ/Gain imbalance, DC offset etc.
  - Apply coefficients to QEC module to improve signal performance
- TX DPD test
  - Based on QEC test, do DPD open/close loop training to get DPD coefficients.
  - Apply coefficient to DPD module to further improve signal performance.
- TX CFR test
  - Based on DPD test, enable CFR to cancel the peaks to improve PA non-linearity.
- RX basic functional test
  - Send single tone or 5G TM waveforms from antenna by equipment.
  - Test tool receives the signal and dump the signal into a file, matlab or other graphic tools can be used to analyze the file
- DAC/ADC performance test
  - The test tool can be used to send different specific signal out to DAC
  - The test tool can dump data sent to DAC, and dump data received by ADC.
  - The test tool can send signal to all the 6 DAC channels and receive from all the 6 ADC channels simultaneously.

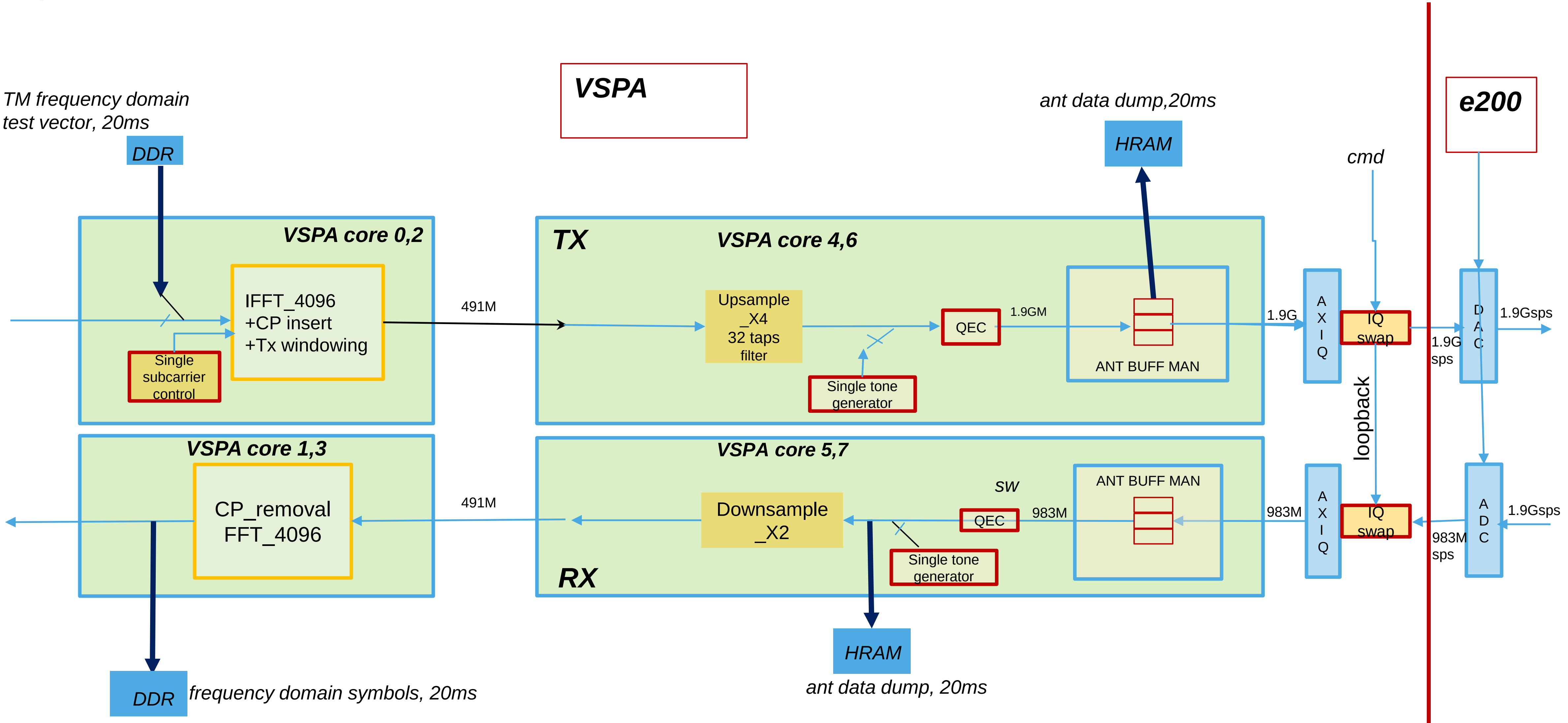
FEATURES

- FR1 FR2 Test Tool is 5G signal generator for RF testing. It supports：
  - LSDCS 4T4R on LA12xx, up to 100Mhz, DAC 491-61Msps, ADC 245-61Msps
  - HSDCS 2T2R on LA12xx, up to 400Mhz, DAC/ADC 1.9Gsps and 983Msps
  - HSDCS 1T1R on LA12xx, up to 800Mhz, DAC/ADC 1.9Gsps and 983Msps
  - FDD 1T1R or TDD 1T2R on LA93xx, up to 25Mhz, DAC 61Msps, ADC 61Msps
  - TDD and FDD for each case
  - Base-station TDD mode and CPE TDD mode.
  - Source waveform can be frequency domain, time domain 1x (option8), time domain 4x (up-sampled), waveform length up to 40ms
  - Time domain single tone (embedded tone generator, no waveform needed), multiple single tones.
  - Frequency domain single sub-carrier
  - IQ SWAP
  - QEC open & close loop training
  - DPD open & close loop training on LS or HS observation path
  - User defined DPD model
  - DPD model auto search
  - TX/RX CFO compensation
  - TX/RX phase compensation
  - Cell tracking
  - SINAD measurement
  - Signal amplitude scaling
  - Memory DMA copy over PCI latency test (LA12xx)
  - Capture/dump TX signal (time domain) or RX signal(time domain & freq domain) up to 20ms

FR1/LSDCS TEST TOOL BLOCK DIAGRAM



FR2/HSDCS TEST TOOL BLOCK DIAGRAM



VERSION CONTROL

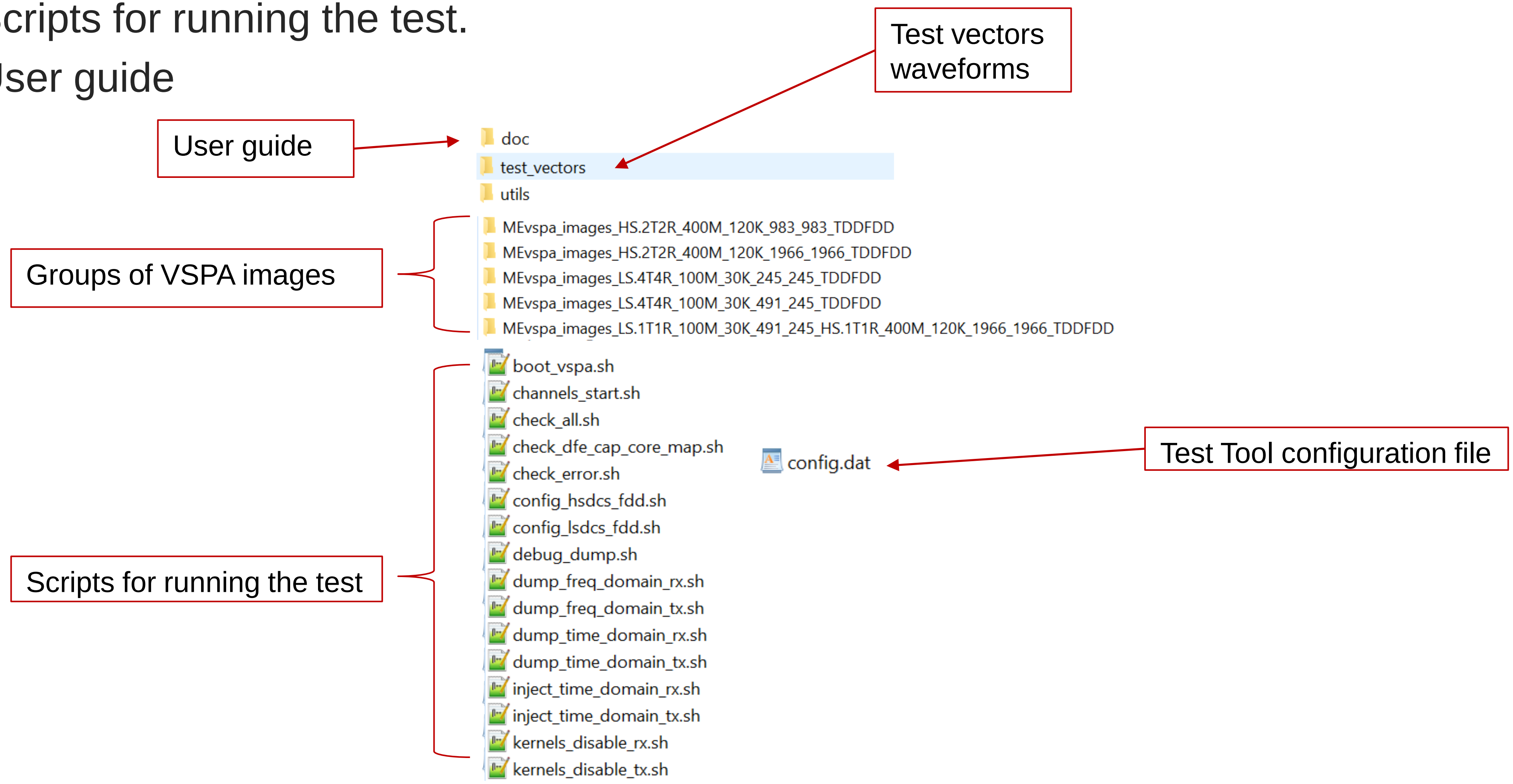
The Test Tool will have 3 levels/versions:

1. **Basic(BA)**: QEC/CFR/DPD disabled
   - All functionalities enabled except QEC/CFR/DPD
   - Released in BSP, for users who just want to check if signal is passing from LA12xx device to RF antennas, who don't want to tune signal quality/performance
2. **Medium(ME)**: CFR/DPD disabled
   - All functionalities enabled except CFR/DPD
   - Release on Flexera, only RU/ISC customers can download through their NXP account
3. **Advanced(AD)**: all functionalities enabled
   - For NXP internal user.
   - Release to customers upon request.
   - For customers who do need to test CFR/DPD and confirm the CFR/DPD parameters are what they need.

The difference in the 3 versions are in VSPA images. Test scripts are same.

WHAT'S IN THE TOOL PACKAGE

- 1. VSPA images
- 2. Scripts for running the test.
- 3. User guide

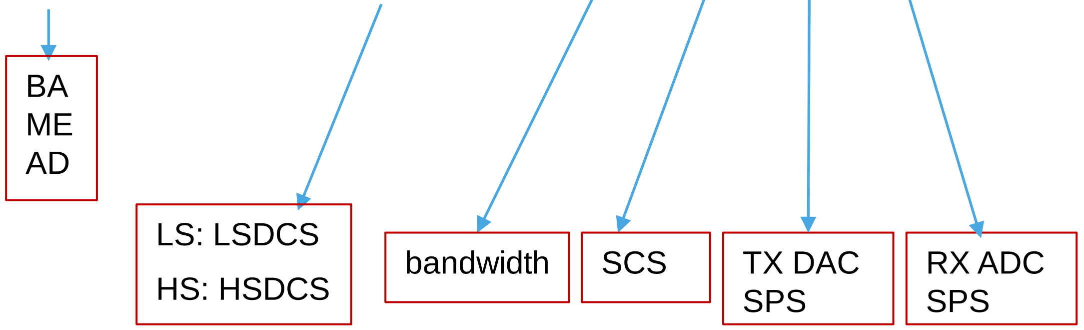


WHAT'S IN THE TOOL PACKAGE

- VSPA images, users should choose desired images to use for the test, each of the following folders contains 8 VSPA images for the 8 VSPA cores. Example VSPA image folder names:
  - MEvspa_images_LS.4T4R_100M_30K_491_245_TDDFDD
  - MEvspa_images_LS.4T4R_100M_30K_245_245_TDDFDD
  - MEvspa_images_HS.2T2R_400M_120K_1966_1966_TDDFDD
  - MEvspa_images_HS.2T2R_400M_120K_983_983_TDDFDD
- Scripts
  - Users just run the scripts to test. All scripts should be run on LX2 console.
- Utils
  - Various tools
- Test vectors
  - TM frequency domain/option8 input test vectors.
- Doc
  - User guide
  - License files

VSPA image folder naming rule:

MEvspa_images_LS.4T4R_100M_30K_491_245_TDDFDD



TEST VECTORS/WAVEFORMS

Images \ input vectors	Default vectors used	
LS_100Mhz	TDD/AD version: TM1.1_100MHz_30kHz_FDD.bin	
	FDD/AD version: TM1.1_100MHz_30kHz_TDD.bin	
HS_400Mhz	TDD: TM3.1_400MHz_120kHz_TDD.bin	
	FDD: TM3.1_400MHz_120kHz_FDD.bin	
HS_800Mhz	FDD: TM3.1_800MHz_480kHz_FDD_option8.bin	

All the test vectors have CELL ID = 1

PREPARE FOR THE TEST

1. Check/change tool configuration file (config.dat)

- The tool is configured by default to:

configurations	Default values	comment	Other possible values and use cases
TDD/FDD	tx_fdd=1 rx_fdd=1		tx_fdd=0 rx_fdd=0 : TDD tx_fdd=0 rx_fdd=1 : TX TDD, RX FDD. If externally loopback TX to RX, RX can dump TX TDD signal
DCS channels to enable	DCSchan=(1 1 1 1 1 1)	All 6 DCS channels enabled	If users want to inject 4x time domain waveform, only 1 channel should be enabled.
TDD pattern	pattern0="7 6 2 4"		
eCPRI compression	ecpri_comp_decomp_enable=0		
option8 or ecpri	option8=0	Freq symbol data is eCPRI.	
IQ SWAP	iqswap_tx=(0 0 0 0 0 0) iqswap_rx=(0 0 0 0 1 0)	IQ SWAP enabled on HSADC0, which is required due to hardware limitation	
DCS enabling	use_nxp_l1c=1	Using NXP provided L1C to enable DCS	Set to 0 if users want to use their own TBGEN code to enable DCS
waveform length	20ms for TDD, 10ms for FDD	Inject 20/10ms and dump 20ms	Can be changed to 1-40ms in need.

- If users want different configurations, edit file config.dat and change the configurations to desired. More detailed descriptions on how to change the config.dat is inside file config.dat

2. Check e200 image (for LA12xx only)

If users are testing FDD mode, they can use default yami.ko and default e200 image (FDD is irrelevant to e200 image).

If users are testing TDD mode, they should use geul_e200_rudemo.elf or their own e200 image which support control of AXIQ/DCS to work in TDD mode

If users rebuild e200 image, they can build with RUDEMO flag ON (RUDEMO=1) to support TDD.

Test tool provides an option to choose which e200 image to use (default is geul_e200_rudemo.elf) by passing e200 image path and name to ./channels_start.sh)

PREPARE FOR THE TEST

- Choose desired VSPA image by test case

Image folder name	Comment	Limitations / test cases
ADvspa_images_LS.4T4R_100M_30K_491_245_TDDFDD	ant_id=0-3	Any of the 4 RX channels can be used as normal RX path, or used as DPD loopback path. Support open/close loop DPD on 4 antennas concurrently 1. Standard 4T4R TX and RX path test 2. 4T DPD test (open/close loop) NOTE: For TDD, 4T4R can work normally. For FDD, 4T or 4R or 2T2R can work normally, 4T4R may not work.
ADvspa_images_LS.2T2R.T4x_100M_30K_491_245_HS.2R_400M_120K_1966_1966_TDDFDD	ant_id=0,1 for FR1 ant_id=4,5 for FR2 RX No CFR	Used to test DPD training, input waveform file must be time domain 4x @491.52Msps, better pre CFRed. The LS RX and HS RX can be used as DPD feedback path. This test case is used to test what is the best ACLR can be achieved. 1. Standard 2T2R TX and RX path test 2. 2T DPD test(open/close loop)
ADvspa_images_LS.2T2R_100M_30K_491_245_HS.2R_400M_120K_1966_1966_TDDFDD	Same as above but CFR can be enabled	Same as above, the difference is that this image use frequency domain waveform or option8 waveform as input waveform. CFR in VSPA is present.
MEvspa_images_HS.1T1R_800M_120K_1966_1966_TDDFDD	ant_id=4 Test FR2 800Mhz	Only time domain 983Msps waveform can be used.
vspa_images_LS.4T4R_100M_30K_245_245_HS.2T2R_400M_120K_983_983_TDDFDD_SINAD vspa_images_LS.4T4R_100M_30K_491_245_HS.2T2R_400M_120K_1966_1966_TDDFDD_SINAD	ant_id=0,1,2,3,4,5 Test SINAD	Only single tone supported. Used for SINAD measurement, single tone can be sent from any of the 6 DACs. SINAD measurement can be done on any of the 2 HSADCs.
MEvspa_images_HS.2T2R_400M_120K_1966_1966_TDDFDD	ant_id=4,5 Test 2T2R on HS	./inject_time_domain_tx.sh will use HRAM as waveform source, size is limited to 0.5ms TXRX buffer mode can be used to inject time domain TX and dump time domain RX, in this case waveform length is limited to not more than 2MB Test case 1. Test HS 1T1R, 2T2R.
ADvspa_images_LS.4T4R_100M_30K_245_245_TDDFDD ADvspa_images_LS.4T4R_50M_15K_122_122_TDDFDD	ant_id=0,1,2,3 Test 4T4R on LS	

If your device part number is HSDCS de-featured, DO NOT start HS channels.

START THE TEST: CONSTELLATION TEST
TWO STEPS TO SEE CONSTELLATION

1ST STEP: BOOT VSPA

./boot_vspa.sh <vspa_image_folder_name> [e200_image_name]

- Users should change the path of yami.ko if their path is different from the one in the script
- The script will automatically copy the VSPA images in the specified folder to /lib/firmware and boot them
- The script will set up DAC/ADC clock automatically according to the naming policy of the image folder
- The script will show boot successful and prompt next step
- e200_image_name: optional, set test tool to use specified e200 image. Otherwise it will use geul_e200_rudemo.elf by default.

2ND STEP: CHANNELS START

./channels_start.sh

- *This step will start all LSDCS and all HSDCS channels, using default configurations in file config.dat.*
- *The script will talk to the VSPA images and know which group of images are used, and load relevant test vectors, enable relevant DCS channels automatically if use_nxp_l1c=1, or wait for users to enable DCS manually if use_nxp_l1c=0*
- *After this step, signal is expected to be sent out to DAC and users can start testing RF signal.*

NOW, TM SIGNAL SHOULD BE SENT TO RF CARD. CHECK SIGNAL!

IF USERS ONLY WANT TO SEND SINGE TONE, GO TO SLIDE ***TIME DOMAIN SINGLE TONE TEST*** DIRECTLY.

EXPECTED SIGNAL WAVEFORM AFTER STEP2

- If everything is working fine, the expected signal waveform can be seen at the outside point of DAC, like what the picture shows.
- If no signal is seen or signal is bad, the reasons could be
  - Test tool issue
  - RF card Hardware issue
  - Test equipment setup issue
- To identify whether it’s test tool issue, follow the instructions on the next 2 slides.

STATUS CHECK AFTER STEP 2

- After step2, the console will print test tool information and channel status information like below.
  - RUNNING/NORMAL means the channel is working correctly. Otherwise if the status is STOPPED, the channels is NOT working.
  - Num symbols sent/received, which should keep increasing. Run ./check_all.sh to check status again to get the latest numbers sent/received
- If status is STOPPED, refer to DCS enabling failure page
- Items to check: SW version, product type, bandwidth&DCSrate, status TX, waveform file name, num running channels, ant_id dcs_id mapping

```
*********************************************************************************
***                                DFE capability
***                     FR1 capabilities:
***                         Product Type:   RU TEST_TOOL
***                                  CFR:   ENABLED, num of PASS = 4
***                                  DPD:   ENABLED @491520 KSPS, DPD CONFIG ID = 4,
num coeff = 28
***                            DPD model:   [ 0 0 5 ][ 1 0 2 ][ 0 1 4 ][ 1 1 4 ][ 0
2 4 ][ 2 2 3 ]
***         DPD 4T4R close loop training:   DISABLED
***                   up-sampling filter:   64 taps, 4x interpolation
***                                TXQEC:   ENABLED
***                                RXQEC:   ENABLED
***                                 iFFT:   ENABLED
***                                  FFT:   ENABLED
***                         eCPRI Decomp:   ENABLED
***                           eCPRI Comp:   ENABLED
***                     SINAD measurement:  DISABLED
***                Channels running time:   hh:mm:ss 0:0:11
*********************************************************************************
```

```
******************************  ANT ID 0: TX on CORE4&0  *********************************************
***                          Bandwidth:   TX: 100 Mhz, SCS 30 Khz
***                   DCS channel used:   TX: FDD LSDCS0-0   491520  KSPS
***          Num symbols sent/received:   TX: 0x000031a9
***                             Status:   TX: RUNNING
***                            TX MODE:   Freq domain waveform @100 Mhz,
filename: ./test_vectors/TM1.1_100MHz_30kHz_FDD.bin
***           tx_allowed current state:   HIGH
***         DCS first enabled timestamp:  tx_allowed 0x8a13efd3
***                        Kernel load:   iFFT       : 9%
***                        Kernel load:   CFR        : 0%
***                        Kernel load:   UP sampling: 20%
***                        Kernel load:   DPD        : 50%
***                        Kernel load:   QEC    6%
***                         CFR status:   OFF NPASS = 0
***                         DPD status:   ON
***                         QEC status:   ON
***                      PNSWAP status:   OFF
****************************************************************************************************
FR1 FR2 Test Tool  Version:   4.0
Current VSPA image Version:   4.0.5

Number of enabled channels TX: 4,   RX: 4
Number of Running channels TX: 4,   RX: 4
MODE: TX FDD, RX FDD,    Basestation

Ant Mapping TX:
ant_id  dcsid      dcs          status
0       0          LSDCS0-0     RUNNING
1       1          LSDCS0-1     RUNNING
2       2          LSDCS1-0     RUNNING
3       3          LSDCS1-1     RUNNING

Ant Mapping RX:
ant_id  dcsid      dcs          status
0       0          LSDCS0-0     RUNNING
1       1          LSDCS0-1     RUNNING
2       2          LSDCS1-0     RUNNING
3       3          LSDCS1-1     RUNNING
```

DUMP TX TIME DOMAIN DATA
MAKE SURE DAC IS SENDING EXPECTED DATA

- This step is very important.
- It can tell whether test tool itself is working correctly or not.

The test tool can dump 20ms TX time domain antenna data before DAC with this script.

./dump_time_domain_tx.sh [ant_id]

- This script will dump 20ms TX ant data to a file, and verify it's md5sum, if the dumped ant data are all expected, the script will print CORRECT, *DAC is sending expected signal*
- Users can use Matlab or test equipment to analyze the data

If status TX is running, and this script prints "CORRECT! CORRECT! CORRECT! ***** Ant data sent to DAC are expected,

It means Test Tool itself is working well and sending correct antenna data out to DAC, users should focus on RF card hardware issue or equipment setup issue if signal is not seen or signal is bad.

Otherwise the test tool does not send correct data to DAC, users should check the LA device/board/BSP version/boot log etc for possible reason, or send all test log to NXP local support for help.

NOTE: if users change the default configurations in config.dat, it may not print CORRECT, users should analyze the dumped data and judge whether the dumped data are expected.

ANT ID & DCS ID

- The test tool VSPA images support 4T4R for LS and 2T2R for HS, and in some images support 2T2R for LS. Channel ID or ant_id is used to identify which 1T1R is used.
  - For 4T4R LS, ant_id is from 0 to 3
  - For 2T2R HS, ant_id is from 4 to 5
  - For 2T2R LS, ant_id is from 0 to 1
  - For 1T1R HS, ant_id is fixed to 4
- There are 6 DCS channels in LA12xx device, DCS ID is used to identify which DCS channel is used.
  - DCS ID 0-3: LSDCS0-3, DCS ID 4-5： HSDCS0-1
- Mapping between ant_id and DCS ID
  - By default, ant_id 0 is mapped to dcs_id 0 and so on.
  - Users can re-map ant_id to other dcs_id if wanted. For example, users is using 2T2R image and want ant_id 0 to map to dcs_id 2.
    run ./channels_start.sh dcs2
  - Ant_id is used in all scripts to identify which channel the script will operation on. Users can find the ant_id dcs_id mapping in the log of channels_start.sh, or run ./check_all.sh which will show the mapping.

BOOT VSPA ARGUMENTS

- For HSDCS, only 1.9Gsps sampling rate is provided in VSPA images. Users can use the following boot vspa argument to test 983Msps.
  - hsdiv2: reduce the HS DCS sampling rate by 2, the following example will set HSDCS sampling rate to 983Mhz.
    - ./boot_vspa.sh MEvspa_images_HS.2T2R_400M_120K_1966_1966_TDDFDD hsdiv2

CHANNELS START ARGUMENTS

- ./channels_start.sh can have arguments to control test case easily without editing configurations in config.dat in most commonly used cases.
- Examples:
  - tdd/fdd: TDD/FDD mode.   txtdd/txfdd/rxtdd/rxfdd: mixed TDD/FDD in TX and RX(only support 1 channel)
    - ./channels_start.sh tdd
    - ./channels_start.sh 0 txtdd rxfdd
  - option8: mode (time domain 1x waveform)
    - ./channels_start.sh option8
  - txonly/rxonly: test only TX or RX on all channels
    - ./channels_start.sh txonly
  - dcsX: test only specified DCS channels
    - ./channels_start.sh dcs2
    - ./channels_start.sh dcs2 dcs4
  - txdcsX, rxdcsX: test only TX or RX on specified DCS channels
    - To send to DCS2 and to receive from DCS3: ./channels_start.sh txdcs2 rxdcs3
  - 1ms: specify TX input waveform length, can be 1ms 10ms 20ms
    - ./channels_start.sh 1ms
  - HRAM: TX waveform put on HRAM. In case PCI bandwidth is not high enough, TX waveform file should be put on HRAM
    - ./channels_start.sh 1ms HRAM
  - waveform_file: filename of user customized TX waveform file. Use this argument to specify the TX waveform file. File size must match with TX waveform length
    - ./channels_start.sh 1ms HRAM user_wv_file.bin
  - mimo: different antennas use different TX waveforms.
    - ./channels_start.sh 10ms mimo
  - cpe: support CPE TDD pattern (Downlink is RX and uplink is TX)
    - ./channels_start.sh cpe

INPUT WAVEFORM

- Test tool supports 3 types of input waveforms with length 1-40 ms
  - Frequency domain input waveform
    - Start channels with ./channels_start.sh
  - Time domain non-upsampled waveform (option8)
    - 122.88Msps for 100Mhz, 491.52Msps for 400Mhz
    - ./channels_start.sh option8
  - Time domain upsampled waveform (upsampled to DAC sampling rate)
    - 491.52Msps for LSDAC, 1.9Gsps for HSDAC (assume LSDAC @491Msps, HSDAC@1.9Gsps)
    - After channels are started, run ./inject_time_domain_tx.sh
- PCI bandwidth requirement if waveform is on host side DDR
  - To support above waveforms, PCI bandwidth requirement is

	Freq domain	time domain non-upsampled	time domain upsampled
LSDAC@491Msps	3Gbps	4Gbps	16Gbps
HSDAC@1.9Gsps	12Gbps	16Gbps	64Gbps

  - If users do not have enough PCI bandwidth, they can choose to use HRAM to hold the waveform file. In this case waveform length is limited to HRAM size which is 6MB.

UPDATE TX TEST VECTOR

- If users have their own frequency domain test vector file, or they want to test other test vector provided within or out of the test tool package, they can run the following script to load it and it will be sent by the test tool immediately.

./update_test_vector.sh [ant_id] <user_vec_filename>

- ***When in mimo mode, use this script to update different waveform files to different antennas***

- If users want to go back to test default TM3.x test vector, run the script again without argument

./update_test_vector.sh

- Example of switching between 400Mhz SCS120 and 100Mhz SCS120 test vector dynamically

```
./boot_vspa.sh <image_folder_name>
./channels_start.sh                //after this script is run, the tool will send default 400Mhz SCS120 waveform.
./update_test_vector.sh  NR-FR2-TM3.1_100MHz_120kHz_TDD_fd_3168_20ms.bin     //after this, 100Mhz SCS120 will be sent
./update_test_vector.sh            //after this script is run, the tool will send default 400Mhz SCS120 waveform
./update_test_vector.sh  NR-FR2-TM3.1_100MHz_120kHz_TDD_fd_3168_20ms.bin     //can switch repeatedly
```

- If users want to make their own test vector, they should follow the following format:
  - Total 20ms length (560 symbols for 100Mhz SCS30Khz, 2240 symbols for 400Mhz SCS120Khz.
  - Each symbol takes 3296 samples(3276 IQ samples+20samples 0 padding) for 100Mhz, 3168 samples for 200/400Mhz
  - Each sample 16bit I + 16bit Q, 2’s complement format, I at lower address, little endian.

SCALE THE SIGNAL AMPLITUDE

- **<u>Scaling output signal</u>**
- In case the signal amplitude is too low or too high, users can run the following script to scale up or down the signal amplitude.
- The script can be run multiple times until the power/amplitude is desired.
- The IQ sample may get saturated, users should check the time domain dump or waveform signal to know whether or how much the IQ data are saturated.
- Output scaling works for all waveforms including frequency domain, option8, injected time domain waveform or single tone (from test tool version 4.0)

./scale_percent.sh [ant_id] [percentage]

➢ *Percentage: any integer number. Such as 110 will scale signal amplitude to 110% on both I and Q*

- **<u>Scaling input signal</u>**
- In case the input data fetched from HighPhy/DU/DDR are too big or too small, users can scale the input up or down to avoid saturation caused by DFE internal data processing.

./scale_percent.sh [ant_id] [percentage] [input]

➢ *For example: ./scale_percnet.sh 0 125 input   will scale input to 125% for antenna 0*

TIME DOMAIN SINGLE TONE TEST

SEND SINGLE TONE

./send_single_tone.sh [ant_id] [freq] [amp] [I|Q] [add] [rx]

- *This script will send singletone to ant_id with specified frequency and amplitude.*
- *freq: frequency in Hz*
- *amp: amplitude of the single tone. 1-100 representing 1%-100% of DAC full scale. Default is 80% if not specified. If amp is 0, DAC will send all 0's. This can be used to stop sending signal on DAC.*
- *I|Q: send on I path only or Q path only, the other path will send all 0.*
- *add: add current single tone on top of previous single tones. Without add, all previous tone will be removed.*
- *For more details please see usage description in the script file*

EXAMPLE: SEND ONE SINGLE TONE OF 10MHZ WITH 80% DAC SCALE

./send_single_tone.sh 0 10000000 80

EXAMPLE: SEND 4 SINGLE TONES SIMULTANEOUSLY, EACH TONE CAN HAVE DIFFERENT FREQ AND AMP

./send_single_tone.sh 0 10000000 20

./send_single_tone.sh 0 20000000 30 add

./send_single_tone.sh 0 30000000 10 add

./send_single_tone.sh 0 40000000 25 add

STOP SINGLE TONE AND GO BACK TO CONSTELLATION TEST

./send_single_tone.sh [ant_id] stop

FREQUENCY DOMAIN SINGLE SUB-CARRIER TEST
SEND SINGLE SUB-CARRIER

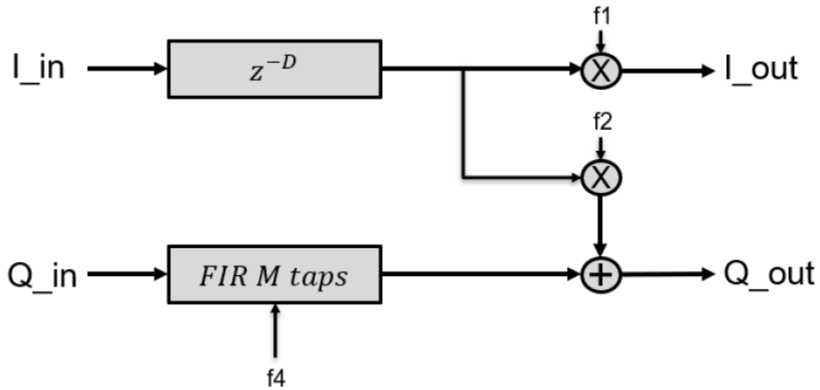
./send_single_subcarrier.sh [ant id] [subcarrier_index] [sample_value] [add]

- *This script will send frequency domain single sub-carrier to ant_id at specified subcarrier index.*
- *Subcarrier index: index of the subcarrier, -2048 to 2047.*
- *sample_value: format: 0xQQQQIIII, 2'complement*
- *Add: add a new sub-carrier on top of previous sub-carriers.*
- *example: ./send_singletone.sh 0 1 0x7FFF7FFF will send 1*(30 or 60 or 120)Khz subcarrier for SCS30/60/120 with sample value 0x7FFF7FFF on ant 0*
- *example: ./send_single_subcarrier.sh 1 100 0x30002000 will send 100*(30 or 60 or 120)Khz subcarrier for SCS30/60/120 with sample value 0x30002000 on ant 1*
- *example: ./send_single_subcarrier.sh 1 stop will stop sending single subcarrier on antenna 1*
- *For more details please see usage description in the script file*

STOP SENDING SINGLE SUBCARRIER

*./ **send_single_subcarrier.sh** [ant_id] stop*

QEC ALGORITHM



- QEC will do the following on TX and RX ant data
  - IQ imbalance compensation
  - Gain compensation
  - DC offset compensation
  - Timing skew compensation (Delay between the I and Q. Not all VSPA images. Currently only ADvspa_images_LS.2T2R.T4x_100M_30K_491_245_HS.2R_400M_120K_983_983_TDDFDD supports with filter taps from 6 to 16)
- QEC coefficient data structure
  - Users should generate QEC coefficients in the following structure, and feed the coefficients to test tool scripts ./update_qec_coeff_tx/rx.sh

```
typedef struct{
    cfixed16_t  state[32];
    cfloat32_tFD_taps[16];
    uint32_t int_del;
    uint32_t FD_tap_cnt;
    float32_t f1;
    float32_t f2;
    float32_t  f4;
    cfloat32_t g;
    cfloat32_t dc;
}qec_params_t;
```

TEST QEC (OPEN LOOP)

Before testing QEC, users should measure IQ imbalance level and calculate coefficients for QEC, imb2qec.py can be used as QEC coeff generation tool. After users get the QEC coefficients file, run the following script to update QEC coefficients, test tool will apply the new coeff immediately.

./update_qec_coeff_tx.sh <ant id> <filename>

- *The script will load the file and send coeff in the file to QEC.*
- *Users should build the same coeff data struct in the file as that in the scipt, little endian.*

If users want to pass through QEC, run the following:

./update_qec_coeff_tx.sh <ant id> dis

- *The script will send pass-through coeff data struct to test tool.*

Users can also manually input the coefficients into the script, and run the script below to update coeff:

./update_qec_coeff_tx.sh [ant id]

- *This script will send qec_user_coeff data struct configured in the script to test tool.*

To test RX QEC, it's the same way, but using script ***./update_qec_coeff_rx.sh***

For close loop QEC, refer to FR1_FR2_test_tool_DPD_QEC_training_guide.pdf

TEST DPD (DPD ALGORITHM)

- The DPD Algorithm

$$-y(n) = \sum_{r,q} \sum_{p=0}^{P(r,q)} \{ x(n-r) * a_{rqp} * |x(n-q)|^{p} \}$$

  - x:input signal, y:output signal, r:delay of signal, q:delay of signal envelope, P:polynomial, a:complex coefficients.
  - r,q,P are integers, can be any numbers. Users should determine r,q,P values according to their PA.
  - An example DPD model (determined r,q,P values) is shown below, which means when r=0,q=0,p=0~6; when r=1,q=0,p=0~3; etc.

- Two DPD models used in test tool.

VSPA images (for DPD training with freq domain or option8 waveforms):

ADvspa_images_LS.2T2R_100M_30K_491_245_HS.2R_400M_120K_983_983_TDDFDD
ADvspa_images_LS.4T4R_100M_30K_491_245_TDDFDD
(NOTE: the actual dpd model in vspa images may be slightly different. Users can see the actual dpd model by ./check_all.sh

r	q	P
0	0	6
1	0	3
0	1	5
1	1	5

VSPA images (for DPD training with 4x up-sampled time domain waveform):

ADvspa_images_LS.1T1R.T4x_100M_30K_491_245_HS.1R_400M_120K_983_983_TDDFDD

- By using this image, users can test customized DPD model by changing P(max) to different values which enables users to find out a best DPD model for their PA.
- When users have changed the DPD model in config.dat at runtime, just run ./dpd_training_close/open_loop.sh to train it immediately without reboot or restart.

r	q	P
0	0	6
1	0	5
2	0	3
3	0	3
4	0	3
5	0	3
0	1	5
1	1	6
2	1	5
3	1	5
4	1	5
5	1	5
0	2	5
2	2	6
0	3	5
3	3	5
0	4	5
4	4	5
0	5	5
5	5	5

refer to FR1_FR2_test_tool_DPD_QEC_training_guide.pdf

TEST DPD (OPEN LOOP, TRAINING BY USERS OWN SOFTWARE)
SKIP THIS PAGE IS USERS USE NXP PROVIDED TRAINING SOFTWARE

The test tool supports open loop DPD by providing (1)script to dump DPD input/output 20ms data, (2)script to update DPD coeff. Users should collect PA output data by equipment, using PA output data and dumped DPD output data to do DPD training and generate DPD coeff, and update coeff to DPD by script.

Steps:

1. Dump DPD output data to a file

./dump_time_domain_tx.sh [ant id] dpdo [num_32KB]

- *The script will dump 20ms (if num_32KB is not specified) DPD input or output data to a file. For more information please refer to the script.*

2. Capture PA output by equipment

3. Do DPD training by users' own software, or by NXP provided software in the test tool, and generate DPD coefficients.

4. Update DPD coefficient from a file

./update_dpd_coeff.sh <ant id> <filename>

- *The script will load the file and send coeff in the file to DPD actuator.*

If users want to set DPD to passthrough mode, run this command:

./update_dpd_coeff.sh <ant id> dis

- *The script will send passthrough coeff to DPD. For more information please refer to the script.*

TEST DPD (OPEN LOOP TRAINING USING NXP PROVIDED TRAINING SOFTWARE)

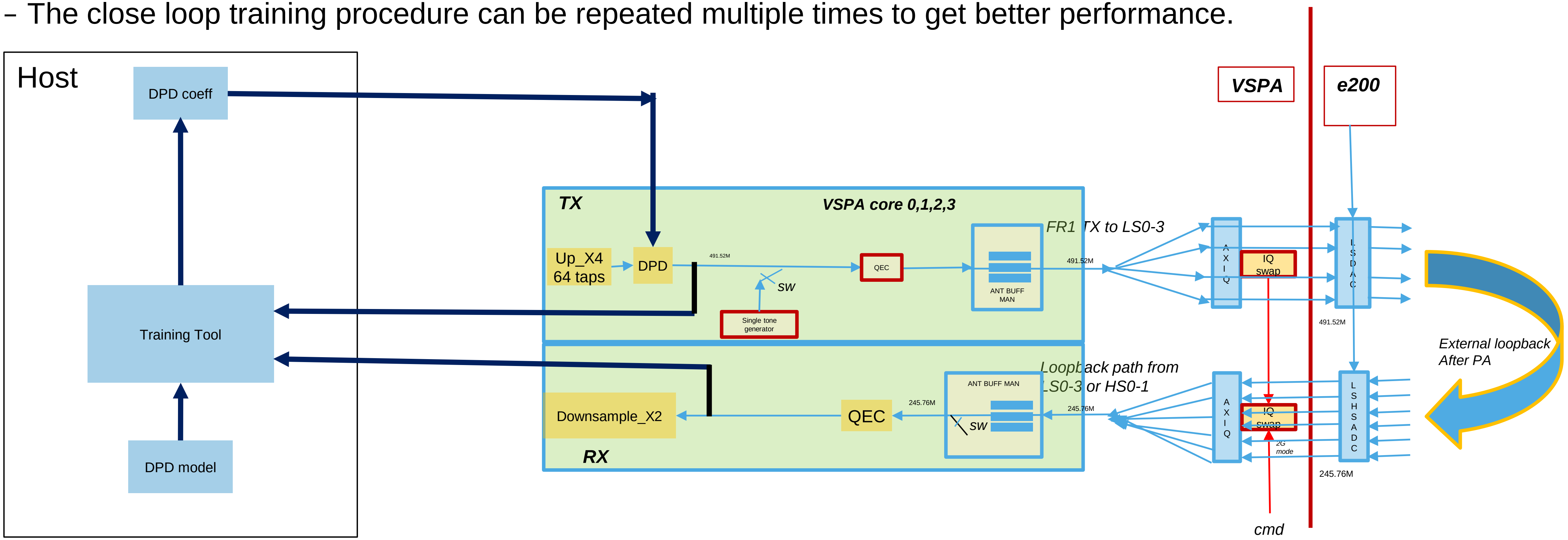
- The Test Tool provides a training software/tool which can train the DPD coefficients with DPD output dump and PA output dump.
  - DPD output dump is captured by the test tool script, length is at least 256KB, format is 16bit 2's complement.
  - PA output dump should be captured by users on their board using equipment, first sample must be aligned to 40ms boundary, duration is at least 256K samples, format is 16bit 2's complement
- Then run the following script to do open loop DPD training, the script includes all the steps illustrated in previous page.

./dpd_training_open_loop.sh <ant_id> <PA_capture_file> <PA_capture_SPS>

- ***PA_capture_sps: sampling rate of the dump, KSPS, example: 491520, 245760, etc***
- ***See the script for details***

DPD CLOSE LOOP TEST

- Test tool can do DPD close loop training on either LS channel or HS channel
  - TX channel can be routed to any of the 4 LS DAC channels.
  - Loopback channel can be from any of the 6 LS+HS ADC channels (it's users' responsibility to control RF card to loopback the TX signal from PA output back to any of the 6 channels).
  - DPD output and feedback dumps are taken to training tool which will generate DPD coefficients according to configured DPD model.
  - DPD coefficients will be updated to DPD actuator.
  - The close loop training procedure can be repeated multiple times to get better performance.



TEST DPD (CLOSE LOOP TRAINING)

- The close loop training script will do the following repeatedly and automatically:
  - Dump DPD output data and feedback data
  - Train DPD coefficients
  - Update DPD coefficients to DPD actuator.
- Users can use either LS or HS RX path for their feedback channel on their hardware, need choose correct vspa images
  - ADvspa_images_LS.2T1R.T4x_100M_30K_491_245_HS.1R_400M_120K_983_983_TDDFDD: support feedback path on LS or HS, and DPD training only on 1 antenna (any one of the 6 LS+HS channels)
  - ADvspa_images_LS.4T4R_100M_30K_491_245_TDDFDD: support feedback path on LS only, support training on 4 antennas.
- Script for close loop training:

./dpd_training_close_loop.sh <ant_id_tx> <ant_id_fb> [training_count] [offset=N]

- ***ant_id_tx: ant_id 0,1,2,3 of LS TX channel to be trained***
- ***ant_id_fd: ant_id 0,1,2,3,4,5 of feedback channel***
- ***training_count: number of times. 1-just train 1 time. Can be any integer number***
- ***Offset=N; Do DPD training with an offset from the slot boundary. For TDD, as signal is not steady at the TDD switching point, do DPD training at a later time is required, this offset specifies from how far from the switching boundary DPD training will be done.***

- example: if users put LSDAC0's feedback signal on LSADC0
  - ***./dpd_training_close_loop.sh 0 0 1***
- example: if users' put LSDAC0's feedback signal on HSADC0
  - ***./dpd_training_close_loop.sh 0 4 1***

- the script can be run multiple times to train multiple times
  - ***./dpd_training_close_loop.sh 0 0 1***
  - ***./dpd_training_close_loop.sh 0 0 1***
  - ***./dpd_training_close_loop.sh 0 0 1***

  Is same as
  - ***./dpd_training_close_loop.sh 0 0 3***

- If DPD training result is not good, ask the following questions:
  - Is TX QEC and RX QEC done before DPD training?
  - Is RFIC calibrated?
  - Is signal quality on feedback path good enough?

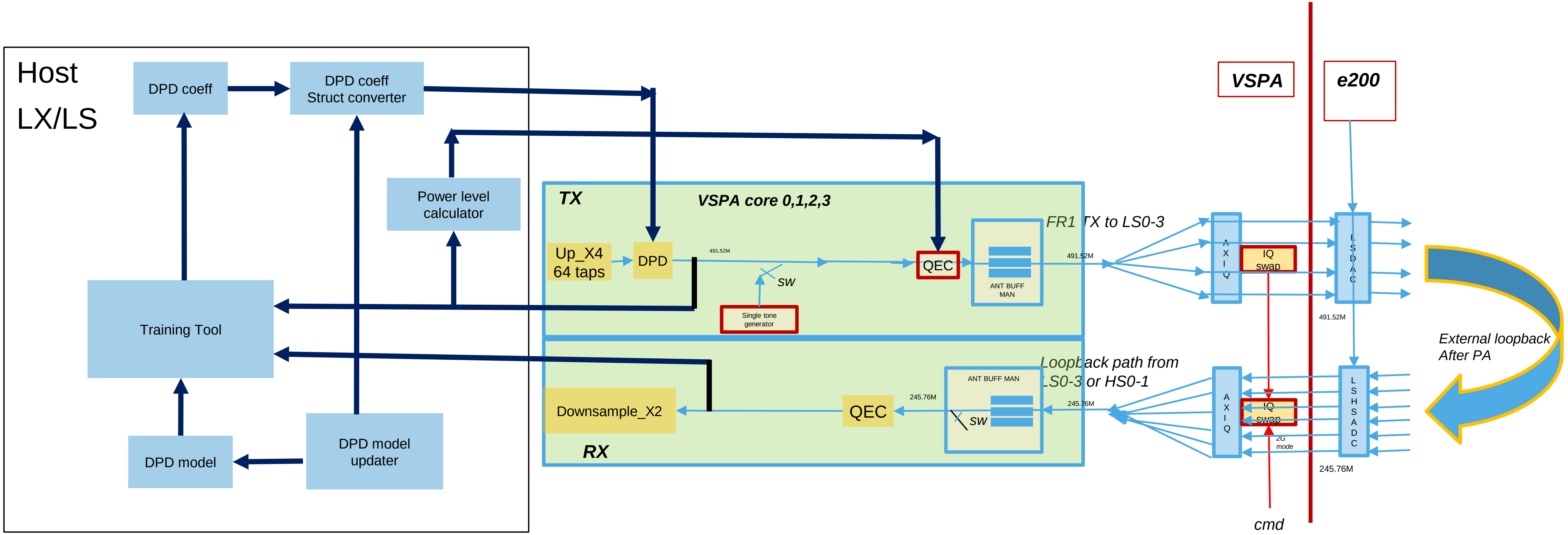
TEST DPD: CUSTOM DPD MODEL

- Big DPD model
  - Test tool has implemented a highly optimized DPD model with 115 coefficients in the red box
  - It supports any customized DPD model with num of coeff equal to or smaller
  - The aim is to enable users to define their own DPD model and test it in test tool, or try different models to see which model is best for their PA
  - See config.dat for how to define customized DPD models
  - Refer to ***FR1_FR2_test_tool_DPD_QEC_training_guide.pdf*** for steps for DPD training
- General steps to test customized DPD models
  - 1. boot vspa using image ADvspa_images_LS.1T1R.T4x_100M_30K_491_245_HS.1R_400M_120K_983_983_TDDFDD
  - 2. channels start
  - 3. do QEC on TX path and observation path
  - 4. do DPD close/open loop training multiple iterations to see the performance
  - 5. change DPD model in config.dat
  - 6. repeat step4 and step5 to see performance of different DPD models

m	q	k(max)
r	q	p(max)
0	0	6
1	0	5
2	0	3
3	0	3
4	0	3
5	0	3
0	1	5
1	1	6
2	1	5
3	1	5
4	1	5
5	1	5
0	2	5
2	2	6
0	3	5
3	3	5
0	4	5
4	4	5
0	5	5
5	5	5

AUTOMATIC DPD MODEL SEARCH

- Test tool can automatically search for best DPD model for current PA or RF card
  - DPD model updater will update DPD model according to predefined policy, currently tens of different DPD models are used.
  - DPD coefficients data struct is converted to match big DPD model required struct.
  - In case a bad DPD model is tried, or in case feedback dump is not a good dump (such as external loopback is not done correctly), the coefficients generated by training tool may cause DPD output signal power to increase to unexpected level and it may damage PA. So before DPD coefficients are applied to DPD actuator, signal output to DAC is shut down in QEC (gain set to 0).
  - Power level calculator will monitor DPD output signal power level, and only when the power level after coefficients are applied is within expected level, it will control QEC to resume the output signal.



AUTOMATIC DPD MODEL SEARCH

- Users can use test tool to automatically search for best DPD model for their board.
  - Just pass argument ‘srch’ in close loop training script
  - Only T4x images support search.
  - The search typically costs around 10 minutes
  - When searching finishes, the best DPD model will be printed and applied
  - A DPD search report file will be generated in ./dpd_training directory, in which all searched DPD model and its ACLR is listed. Users can pick one that is good for their board.

```
./dpd_training_close_loop.sh <ant_id_tx> <ant_id_fb> [training_count]  [srch]
```

TEST CFR

- Users should configure in config.dat how many CFR PASS they need and in which PASS's sub sampling filter is enabled. Default configuration is CFR disabled.

- Users can enable all CFR

```
./ kernels_disable_tx.sh  [ant_id] cfr –r
```

- Users can disable all CFR PASS's or each PASS

```
./ kernels_disable_tx.sh  [ant_id] cfr
```

- Users can update parameters for all CFR PASS's from a file (users should build the file with struct defined in the script)

```
./ update_cfr_para.sh [ant_id] filename
```

TEST CFO

- To set carrier frequency offset

```
./update_fine_cfo_nco_freq.sh <ant_id> <freq_Hz> [tx|rx]
Example:
./update_fine_cfo_nco_freq.sh 0 1000000        will set carrier offset to -1000000Hz on ant 0 TX
./update_fine_cfo_nco_freq.sh 0 -1000000 rx    will set carrier offset to 1000000Hz on ant 0 RX
```

TEST PHASE COMPENSATION

- Phase compensation is enabled by default by working in passthrough mode (coefficients are configured with all 1.0 which should be updated by host)

- To update phase compensation coefficients:

./update_phase_compensation_coeff.sh <ant id> <coeff_bin_file|addr|arfcn=> [dis] [tx|rx]

Example:

./update_phase_compensation_coeff.sh 0 arfcn=1   will update coefficients for arfcn index 1 on ant 0 TX

./update_phase_compensation_coeff.sh 0 <bin_file> rx   will update coefficients from bin_file on ant 0 RX

Number of coefficients in the bin file:

7 for SCS 15Khz

14 for SCS 30Khz

28 for SCS 60Khz

56 for SCS 120Khz

For each coefficient, the format is 32-bit I and 32-bit Q, float, I at lower address, little endian.

POWER MEASUREMENT

- TX samples power is measured at after QEC before DAC.
- RX samples power is measured at after QEC after ADC
- Samples power is shown in ./check_all.sh log, example:
  - ****************************** ANT ID 0: TX+RX on CORE0 ***
  - *** Bandwidth: TX: Max 20 Mhz, sub-carriers 612, SCS 30 Khz
  - *** Bandwidth: RX: Max 20 Mhz, sub-carriers 612, SCS 30 Khz
  - *** DCS channel used: TX: FDD DAC0 61440 KSPS, RX FDD ADC0 61440 KSPS, HW_2x_Decimation OFF
  - *** Num symbols sent/received: TX: 0x46a2760a ~28/ms, RX: 0x46a276a3 ~27/ms
  - *** Symbol Type: TX: OPTION7-2, RX: OPTION7-2
  - *** Sample power at DAC: TX: 0.052918/sample, -12.8 dB, max I=-0.876953, max Q=-0.889648
  - *** Sample power at ADC: RX: 0.000887/sample, -30.5 dB, max I=0.973633, max Q=-0.030762
  - *** TX MODE: Freq domain waveform

AXIQ INTERNAL DIGITAL LOOPBACK

- In HSDCS FDD mode, users can enable AXIQ digital loopback to loopback TX time domain antenna data back to RX time domain.
- Loopback script should be put after ./boot_vspa.sh and before ./channel_start.sh

```
./boot_vspa.sh <image_folder_name>
./channels_start.sh loopback
```

- NOTE: loopback is not supported in LSDCS

TEST 4X UP-SAMPLED TX TIME DOMAIN WAVEFORM (LS ONLY)

- 4x time domain waveform: up-sampled to 491Msps. Supported only by T4x images
- Only VSPA images tagged with T4x support 4x time domain waveform

- To send 4x time domain waveform, run

```
./boot_vspa.sh <vspa_folder_name_tagged_with_T4x>
./channels_start.sh
```

TEST OPTION 8

- Option 8 test allows users to play time domain test vector for both LS and HSDC
  - Set option8=1 in config.dat
  - The test tool provides default option8 time domain test vectors for LSDCS 100Mhz and HSDCS 800Mhz only. In case users run option8 for other cases, the script will prompt users to load their own option8 vectors.
- Steps to test option8

./channels_start.sh option8 *//after this step the tool is sending default option8 vector for LSDSC*

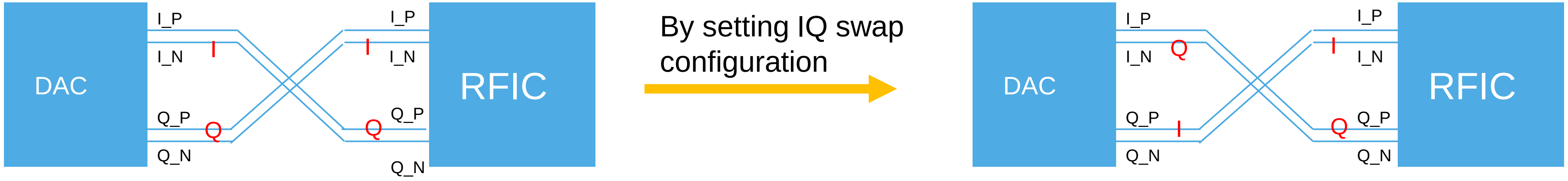
./update_test_vector.sh [ant_id] *<option8_vec_filename> //run this step to change to user option8 test vector.*

- Difference of testing option8 and testing time domain waveform
  - Testing time domain waveform is injecting 4x up-sampled waveform file directly to DCS without any software processing except QEC. All other scripts will have no effect.
  - Testing option8 is injecting 1x time domain waveform (not up-sampled) to the DFE data flow, the waveform will go through CFR, DPD,up-sampling and QEC. Users can run all other scripts to process the waveform.

TEST IQ SWAP (FOR HARDWARE MISTAKES)

- If users made mistake in their hardware design which swaps I and Q on DAC output signal, they need test IQ swap. IQ swap is done by VSPA to configure AXIQ control register, the hardware will swap IQ according to register value.
- For other users, they do not need test IQ swap
- To test IQ SWAP
  - Change IQ SWAP configurations in config.dat, see below red box.
  - Reboot and start the test
  - Users need equipment to measure the DAC output signal to verify IQ SWAP

```
#IQ SWAP configurations: All TX/RX for all 6 DCS channels are NOT swapped.
#The 6 values in each parentheses are for the 6 DCS channels, from left to right mapping to
LSDCS0-0,LSDCS0-1,LSDCS1-0,LSDCS1-1, HSDCS0, HSDCS1
#To enable IQSWAP, change the value to 1
#example: iqswap_tx=(0 0 1 0 0 0) will enable IQ SWAP on TX of LSDCS1-0
iqswap_tx=(0 0 0 0 0 0)
iqswap_rx=(0 0 0 0 0 0)
```



Wrong hardware connections

LA device swaps I and Q, signals on RFIC get correct

IQ SWAP is zero cost.

TX TEST GENERAL CASE

– send frequency domain TM waveform/constellation:
  - ./boot_vspa.sh <image_folder_name>
  - ./channels_start.sh

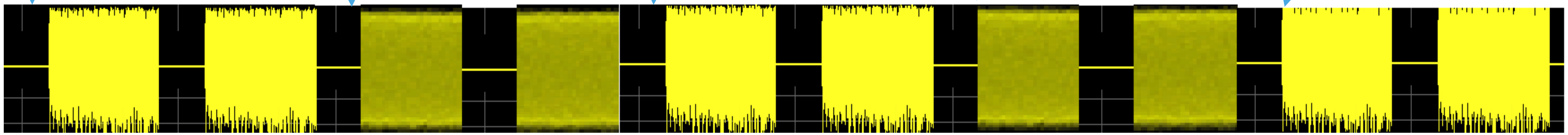
- ./send_single_tone.sh 0

- ./send_single_subcarrier.sh 0

- ./send_single_tone.sh 0 stop

- ./send_single_subcarrier.sh 0 stop

1st step for all test



Dumps can be taken at any time during the test.

TEST RX: DUMP TIME DOMAIN RX ANTENNA DATA

```
./dump_time_domain_rx.sh [ant_id] [num_32KB]
```

- Users can run above script to dump RX antenna data to a file
  - The length of dump is ***num_32KB*32KB,*** if ***num_32KB*** is not specified the length is 1-40ms defined by dump_time_len in config.dat.
  - Use Matlab to analyze the spectrum, constellation, EVM of RX ant data.
  - The antenna data is captured from right after the DAC/AXIQ.
  - If QEC is enabled, the dump is QECed.
  - Dump data format: 2's complement, 16bit I Q. little endian, I is at lower address.

- Dump multiple RX antennas at the same time
  - In case for MIMO, users can dump multiple antennas at the same time.

```
./dump_time_domain_rx.sh [ant_id] [+ant_id  +ant_id  ...] [num_32KB]
```

- Example:
  - ./dump_time_domain_rx.sh 0 +1 +2 +3 1   will dump ant 0,1,2,3 at the same time with size 32KB

TEST RX: DUMP RX FREQUENCY DOMAIN SYMBOL DATA

```
./dump_freq_domain_rx.sh [ant_id]
```

- Users can run above script to dump 20ms RX frequency symbol data to a file
  - Use Matlab to analyze the spectrum, constellation, EVM.
  - Dump data format: 2's complement, 16bit I Q. Little endian. I is at lower address.
  - The dump file contains 20ms, 560 symbols for 100Mhz bandwidth, 2240 symbols for 400Mhz bandwidth
  - For 100Mhz, each symbol size is 3296 samples (3276 sub-carriers plus 20 zero-padding for alignement)
  - For 200/400Mhz, each symbol size is 3168 samples.

- NOTE: If option8 is configured, dumping frequency domain will actually dump time domain 1x option8 waveform.

TEST RX: CELL TRACKING

./set_dfe_state.sh track=a:b:c:d:e
-a,b,c,d,e stand for ssb_period,ssb_sym_id,ssb_re_offset,nid2,nid1

– Cell tracking result will be printed, example:
  - ./set_dfe_state.sh track=3:3:0:2:0
    - send_celltrack_cmd 0 3 3 0 2 0
    - !! Cell Tracking request sent. Cell Tracking result: !!
    - cell_id : 0x00000002 = 2
    - time_ofst : 0x00426000 = 66 samples @ 245760KSPS
    - cfo : 0x00002bd6 = 642 Hz
    - ss_rssi_db: 0x0000f644 = .96197509 dB
    - ss_rsrp_db: 0x0000bbe0 = .73388671 dB
    - ss_rsrq_db: 0x000a04a4 = 10.01812744 dB
    - ss_sinr_db: 0x000af254 = 10.94659423 dB

CHECK STATUS

- At any time, or users think the tool is not working well, users can run this script to know the tool status.
  - The script will show what type of images are using, the tool is in running or stopped state, single tone or constellation mode.
  - The script will check if there are any errors reported by the tool and print error information. Once there are errors, users should reboot and re-start the test. Run this script after each step and find out which step causes the error. Send all the log information to NXP support for help in need.

```
./check_all.sh
```

TESTING DIFFERENT DCS CHANNELS

- To test different DCS channels, use different ant_id in the script
  - For example, test single tone on antenna 1, run the following script.

```
./send_single_tone.sh 1
```

- To only enable specific DCS channels, change default configurations in config.dat
  - DCSchan=(1 1 1 1   1 1)   //0:disable, 1:enable

TEST VSPA DMA PERFORMANCE

./measure_dma_latency.sh [dma_size]

- Users can run above script to test VSPA DMA copy performance after boot and before ./channels_start.sh
  - The test will use 1 DMA channel and 2 DMA channels to do the DMA copy respectively.
  - The test will test DDR, PEBM, HRAM respectively. The test will test read and write performance respectively
  - The test will test DMA size 16KB by default, if dma_size is not passed in the command. [dma_size] must be no more than 64KB
  - If users run the script after ./channels_start.sh, the performance result may be impacted by the test tool functionalities.
  - The following is an example of the performance test result on Bonnyrig board:
    - VSPA DMA Memory read/write performance test result:
    - DDR READ by VSPA DMA 1 channel , 16384 bytes, cycle count 3861, 6288 ns, 20 Gbps
    - DDR READ by VSPA DMA 2 channels, 16384 bytes, cycle count 1864, 3035 ns, 43 Gbps
    - DDR WRITE by VSPA DMA 1 channel , 16384 bytes, cycle count 1503, 2447 ns, 53 Gbps
    - DDR WRITE by VSPA DMA 2 channels, 16384 bytes, cycle count 1430, 2328 ns, 56 Gbps
    - PEBM READ by VSPA DMA 1 channel , 16384 bytes, cycle count 741, 1206 ns, 108 Gbps
    - PEBM READ by VSPA DMA 2 channels, 16384 bytes, cycle count 471, 767 ns, 170 Gbps
    - PEBM WRITE by VSPA DMA 1 channel , 16384 bytes, cycle count 669, 1089 ns, 120 Gbps
    - PEBM WRITE by VSPA DMA 2 channels, 16384 bytes, cycle count 639, 1040 ns, 125 Gbps
    - HRAM READ by VSPA DMA 1 channel , 16384 bytes, cycle count 723, 1177 ns, 111 Gbps
    - HRAM READ by VSPA DMA 2 channels, 16384 bytes, cycle count 485, 789 ns, 165 Gbps
    - HRAM WRITE by VSPA DMA 1 channel , 16384 bytes, cycle count 621, 1011 ns, 129 Gbps
    - HRAM WRITE by VSPA DMA 2 channels, 16384 bytes, cycle count 534, 869 ns, 150 Gbps

DCS ENABLING

- By default, DCS is enabled by e200 images from BSP release.
- If users have their own yami and e200 images that they want to use to enable DCS, they can
  - set use_nxp_l1c=0 in config.dat
  - Or ./channels_start.sh nl1c

TEST DIFFERENT TDD PATTERN

- The default TDD pattern configured in config.dat is 7 6 2 4 0 0
- Users can change this configuration to any desired pattern
  - Pattern format: DL slots, DL symbols, UL slots, UL symbols, G slots after DL, G slots after UL
  - Example 1:
    - pattern DDDDD DDSUU   (S=6 DL symbols, 4 GP symbols, 4 UL symbols)
    - Configure the pattern to 7 6 2 4 0 0
  - Example 2:
    - pattern DDDSU   (S=2 DL symbols, 10 GP symbols, 2 UL symbols)
    - Configure the pattern to 3 2 1 2 0 0
- After the change, reboot and re-start.

SINAD TEST/MEASUREMENT

- Test tool can do SINAD measurement, the measurement is triggered by a host command. Upon receiving the command, test tool will do the following in VSPA.
  - Take 32KB (8K samples) of ant data from RX AXIQ
  - Do FFT on the 8K samples (separately on I and Q)
  - In the 8K samples FFT output, add up specified num of samples from specified sample as signal power, and add up the rest samples as noise power(ignore the first a few num of samples which are DC).
  - Calculate SINAD values for I and Q: 10*log10(signal power/noise power)
  - Write SINAD values to specified address.

- Script to trigger SINAD measurement

./sinad_test.sh <ant_id> <start_point> <num_points> [dest_address]

> - *start_point: signal start point in the 8192 FFT output, range 0-8191*
> - *num_points: number of signal points from start point.*
> - *dest_address: the address in host view(40bit) where SINAD result(24bytes needed) are stored, must be 4KB aligned. must be on DDR. if not specified, test tool will assign the address*
> - *example: ./sinad_test.sh 4 1660 13   will test SINAD for antenna 4, signal starts from point 1660 and end at point 1672 total 13 points(400Mhz single tone at 1.9Gbps)*

- The following information will be printed on console (values are just for example):
  - SINA measurement result:
  - Noise Win Start = 10
  - Noise Win End   = 3332
  - Signal Start   = 2910
  - Signal Len   = 13
  - SINAD_I at address: 0x236429e000, Q at address 0x236429e004, data type is 32-bit float little endian.
  - SINAD_I=0x42796000
  - SINAD_Q=0x42796000

SINAD TEST/MEASUREMENT

- Required vspa images:
  - vspa_images_LS.4T4R_100M_30K_491_245_HS.2T2R_400M_120K_1966_1966_TDDFDD_SINAD
  - vspa_images_LS.4T4R_100M_30K_245_245_HS.2T2R_400M_120K_983_983_TDDFDD_SINAD
- Required mode
  - FDD (set tx_fdd=1, rx_fdd=1 in config.dat)
- Required steps
  - ./boot_vspa.sh <vspa_image_folder>
  - ./channels_start.sh sinad
  - ./send_single_tone.sh <ant_id> <freq> <amp>
  - ./sinad_test.sh <ant_id> <start_point> <num_points> [dest_address]
- Required hardware setup
  - Test tool will enable 6T6R, users must externally loopback any one of the 6 TX to any one of the 2 HS RX.
  - Then use above steps to test SINAD on that RX.
  - Example: If HSDAC0 TX is externally looped back to HSADC1 RX, use the following steps to test 400Mhz/700Mhz at 1.9Gsps
    - ./boot_vspa.sh vspa_images_LS.4T4R_100M_30K_491_245_HS.2T2R_400M_120K_1966_1966_TDDFDD_SINAD
    - ./channels_start.sh sinad
    - ./send_single_tone.sh 4 400000000 85
    - ./sinad_test.sh 5 1660 13 0x236429e000
    - ./send_single_tone.sh 4 700000000 85
    - ./sinad_test.sh 5 2910 13 0x236429e000
  - If software can control the external loopback from any TX to any RX, ./send_single_tone.sh and ./sinad_test.sh can be run on any and_id repeatedly without reboot or restart.

CHANNEL STATUS

- Number of channels
  - Num of channels supported is decided by VSPA images, the image folder names show the max num of channels supported by the VSPA images, such as LS.4T4R or LS.2T2R.
  - The channels are designated with antenna ID starting from 0.
- Mapping channels to DCS
  - Each channel can be mapped to any DCS ID. The mapping can be done by ant remapping in config.dat, or by passing arguments to channels_start.sh. Example: ./channels_start.sh dcs2 will map ant0 to DCS2. ./channels_start.sh dcs0 dcs2 will map ant0/1 to DCS0/2
- Channels status
  - Uninitialized state: There are 6 DCS channels, for those not started channels, they will stay in uninitialized state. In this state, AXIQ is not initialized, software is not sending any data to AXIQ/DCS. Uninitialized channels are not shown in check_all.sh log
  - Running state: channels are started, the channel is sending/receiving signals (waveform or single tone).
  - Running IDLE state: channels are in running state but is sending 0 to DCS.
  - Stopped state: channels are stopped because of tx/rx_allowed is not enabled, or because of errors.
- Switching between channels
  - Users can start all channels to idle state, and enable and test one channel at a time, example:
  - ./boot_vspa.sh ADvspa_images_LS.4T4R_100M_30K_491_245_TDDFDD
  - ./channels_start.sh idle //all channels start as idle
  - ./channels_act_tx.sh 0   //set channel 0 to active, other channels idle
  - ./channels_act_tx.sh 1 //set channel 1 to active, other channels idle
  - ./channels_act_tx.sh all //set all channels active

SPECIAL NOTE

- The tool supports all 4 LSDCS channels simultaneous test
- The tool supports all 2 HSDCS channels simultaneous test
- The tool supports both LSDCS and HSDCS channels simultaneous test

- Special NOTE:
  - As TX/RX dumping increase PCIe and VSPA core internal data throughput a lot, in multi-channel testing, it could incur errors.
  - If users see errors printed after taking dumps, do the following:
    - Reboot VSPA images, re-do the test, do not take dumps, only observe signals on equipment
    - Change config.dat, just enable desired DCS channel, keep other channels disabled, this will reduce a lot of PCIe bandwidth. reboot and try taking dumps again. If there is still errors, do not take dumps.
  - Stop testing after errors occur
    - When test tool shows TX AXIQ underlow, RX AXIQ overflow, or any other error messages, stop testing and fix the errors first, because the system is no longer synchronized with air interface, data may be wrong.

5G NR TEST MODEL WAVEFORM GENERATION

- Matlab 5G NR-TM and FRC Waveform Generation

  [5G NR-TM and FRC Waveform Generation - MATLAB & Simulink (mathworks.com)](mathworks.com)

- Commands and Parameters

  DL:

  dlrefwavegen = hNRReferenceWaveformGenerator(dlnrref,bw,scs,dm,ncellid,sv)

  [dlrefwaveform,dlrefwaveinfo,dlresourceinfo] = generateWaveform(dlrefwavegen);

  dlrefwaveform : DL time domain samples

  dlrefwaveinfo.ResourceGridBWP(or ResourceGridInCarrier): DL freq domain samples

  UL:

  ulrefwavegen = hNRReferenceWaveformGenerator(ulnrref,bw,scs,dm,ncellid)

  [ulrefwaveform,ulrefwaveinfo,ulresourceinfo] = generateWaveform(ulrefwavegen);

  ulrefwaveform : UL time domain samples

  ulrefwaveinfo.ResourceGridBWP(or ResourceGridInCarrier): UL freq domain samples

5G NR TEST MODEL WAVEFORM GENERATION

- Time/Freq domain samples amplitude adjust

  For example,

  Reducing freq domain power to avoid VSPA IFFT saturation

  Increasing time domain power to full DAC range

- Freq domain symbol padding

  In case freq domain symbol size is not vspa dmem line(128Byte) aligned, padding zeros at symbol tail is needed

- Save bin file

  I followed by Q, int16, 2's complement

NEW IN V4.9 VS 4.8

- New VSPA images to support Narrow band:
  - Cell tracking support in both LA12xx and LA9310.

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